

Performance at Noise Level $\sigma = 0$

This is the **ideal, noise-free scenario**, which serves as our baseline.

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.19 ± 0.00	0.46 ± 0.00	0.0002 ± 0.0000
-0.9443	1.33 ± 0.00	0.94 ± 0.00	0.0012 ± 0.0000
0.1559	0.07 ± 0.00	0.16 ± 0.00	0.0001 ± 0.0000
-0.6195	0.79 ± 0.00	0.62 ± 0.00	0.0003 ± 0.0000
-0.8419	1.06 ± 0.00	0.84 ± 0.00	0.0005 ± 0.0000

Performance at Noise Level $\sigma = 0.001$

Here, a **very small amount of noise** is introduced.

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.19 ± 0.00	0.46 ± 0.00	0.0030 ± 0.0006
-0.9443	1.30 ± 0.01	0.94 ± 0.00	0.0028 ± 0.0004
0.1559	0.08 ± 0.00	0.16 ± 0.00	0.0024 ± 0.0004
-0.6195	0.85 ± 0.01	0.62 ± 0.00	0.0029 ± 0.0003
-0.8419	1.08 ± 0.00	0.84 ± 0.00	0.0033 ± 0.0005

Performance at Noise Level $\sigma = 0.01$

The noise is increased by a factor of 10. The controller's performance **degrades dramatically**.

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	N/A	1.57 ± 0.00	N/A
-0.9443	N/A	1.58 ± 0.00	N/A
0.1559	N/A	1.57 ± 0.00	N/A
-0.6195	N/A	1.57 ± 0.00	N/A
-0.8419	N/A	1.57 ± 0.00	N/A

Performance at Noise Level $\sigma = 0.1$

At this **high noise level**, the controller continues to fail.

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	N/A	1.57 ± 0.00	N/A
-0.9443	N/A	1.57 ± 0.00	N/A
0.1559	N/A	1.57 ± 0.00	N/A
-0.6195	N/A	1.58 ± 0.00	N/A
-0.8419	N/A	1.58 ± 0.00	N/A

Analysis of Results

This analysis reveals a classic trade-off in control systems: the controller is well-tuned for an ideal environment but shows a **lack of robustness** as sensor noise increases.

1. Baseline Performance ($\sigma = 0$)

In the noise-free environment, the controller performs exceptionally well.

- **Consistency:** The standard deviation for all metrics is zero, indicating that the system is deterministic and its behaviour is perfectly repeatable.
- **Accuracy:** The steady-state error is extremely low (on the order of 10^{-4} radians), showing the controller can precisely reach and hold the target position.
- **Responsiveness:** The stabilisation time is directly related to the initial angle—the further the pendulum starts from the centre, the longer it takes to stabilise, which is expected behaviour. The overshoot is minimal, essentially equal to the starting angle, indicating a critically damped or slightly underdamped response with no significant oscillation.

2. Impact of Low Noise ($\sigma = 0.001$)

Even a tiny amount of noise has a noticeable effect, primarily on precision.

- **Steady-State Error:** This is the most affected metric. The average steady-state error **increased by an order of magnitude** (from ~ 0.0005 to ~ 0.0030 rad). The non-zero standard deviation shows that the system is no longer perfectly predictable. This happens because the controller is constantly reacting to small, spurious noise signals, causing it to "jitter" around the setpoint instead of settling perfectly.
- **Overshoot and Stabilisation Time:** These metrics remained largely unaffected. The controller's initial, large-scale response to the angle error is strong enough to be unperturbed by such low-level noise.

3. Critical Failure at Higher Noise Levels ($\sigma = 0.01$ and $\sigma = 0.1$)

The system experiences a **total failure to stabilise** once the noise reaches a standard deviation of 0.01.

- **Loss of Stability:** The controller never meets the stabilisation criteria (staying within ± 0.1 rad for 1 second). The noisy signal, particularly when fed into the Derivative (D) term of the PID controller, likely causes erratic and high-magnitude control actions. This makes the cart oscillate or move unpredictably, preventing the pendulum from settling.
- **Massive Overshoot:** The overshoot values (~ 1.57 rad, or ~ 90 degrees) are enormous compared to the initial angles. This indicates that the noisy control signal is so poor that it actively pushes the pendulum far beyond its starting point, leading to instability. The system is essentially flailing uncontrollably.

Conclusion

The PID controller with the gains $K_p=44$, $K_i=80$, $K_d=6$ is **highly optimised for a noise-free environment** but is **extremely sensitive to sensor noise**. While robust enough to handle minuscule noise ($\sigma = 0.001$) with only a slight loss of precision, its performance collapses completely at a noise level of just $\sigma = 0.01$.

This demonstrates a critical concept in real-world control engineering: a controller that works perfectly in simulation can fail dramatically in practice if it's not designed to be **robust** to real-world imperfections like sensor noise. To improve this system, one would need to implement a filter (like a Low-Pass Filter or a Kalman Filter) on the sensor input or re-tune the PID gains (likely by reducing the Derivative gain, K_d) to make it less reactive to high-frequency noise.